

VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



MACHINE LEARNING (CO3117)

Homework 6a:

Bayesian Belief Network

Team LHPD2

Semester 2, Academic Year 2024 - 2025

Teacher:	Nguyen An Khuong	
Students:	Nguyen Quang Phu	- 2252621 (Leader)
	Nguyen Thanh Dat	- 2252145 (Member)
	Pham Huynh Bao Dai	- 2252139 (Member)
	Nguyen Tien Hung	- 2252280 (Member)
	Nguyen Thien Loc	- 2252460 (Member)

HO CHI MINH CITY, FEBRUARY 2025

Contents

1	Introduction	2
1.1	Purpose of This Document	2
1.2	Workload	2
2	Problem Description - Our Solution	3
2.1	Question 6.6	3
2.1.1	Description	3
2.1.2	Solution	5

Chapter 1

Introduction

1.1 Purpose of This Document

This chapter serves as an introduction to our team, LHPD2, for the Machine Learning (CO3117) course in Semester 2, Academic Year 2024 - 2025. The purpose of this writing is to formally present our team members and confirm our participation in solving the given homework exercise, “Homework 6a: Bayesian Belief Network”.

1.2 Workload

Specifically, our contributions included:

- **Understanding the Problem:** Each member participated in analyzing and discussing the given exercise to ensure a shared understanding of the requirements.
- **Exploring Concepts:** We collectively researched and reviewed relevant concepts, theories, and methods necessary for solving the problems.
- **Reasoning for the Solution:** The team worked together to identify the rationale behind each approach, ensuring that all steps were logical and well-justified.
- **Implementing the Solution:** The solutions were implemented with equal collaboration, where every member contributed to coding, calculations, and documentation.
- **Final Review:** The team jointly reviewed the solutions to verify their correctness, clarity, and coherence before submission.

We believe that this exercise has enhanced our understanding of the concepts involved and strengthened our ability to work collaboratively as a team.

Chapter 2

Problem Description - Our Solution

2.1 Question 6.6

2.1.1 Description

Draw the Bayesian belief network that represents the conditional independence assumptions of the naive Bayes classifier for the PlayTennis problem of Section 6.9.1. Give the conditional probability table associated with the node Wind.

Day	<i>Outlook</i>	<i>Temperature</i>	<i>Humidity</i>	<i>Wind</i>	<i>PlayTennis</i>
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

TABLE 3.2
Training examples for the target concept *PlayTennis*.

Figure 2.1

6.9.1 An Illustrative Example

Let us apply the naive Bayes classifier to a concept learning problem we considered during our discussion of decision tree learning: classifying days according to whether someone will play tennis. Table 3.2 from Chapter 3 provides a set of 14 training examples of the target concept *PlayTennis*, where each day is described by the attributes *Outlook*, *Temperature*, *Humidity*, and *Wind*. Here we use the naive Bayes classifier and the training data from this table to classify the following novel instance:

$\langle \text{Outlook} = \text{sunny}, \text{Temperature} = \text{cool}, \text{Humidity} = \text{high}, \text{Wind} = \text{strong} \rangle$

Our task is to predict the target value (*yes* or *no*) of the target concept *PlayTennis* for this new instance. Instantiating Equation (6.20) to fit the current task, the target value v_{NB} is given by

$$\begin{aligned} v_{NB} &= \operatorname{argmax}_{v_j \in \{\text{yes}, \text{no}\}} P(v_j) \prod_i P(a_i | v_j) \\ &= \operatorname{argmax}_{v_j \in \{\text{yes}, \text{no}\}} P(v_j) \quad P(\text{Outlook} = \text{sunny} | v_j) P(\text{Temperature} = \text{cool} | v_j) \\ &\quad P(\text{Humidity} = \text{high} | v_j) P(\text{Wind} = \text{strong} | v_j) \quad (6.21) \end{aligned}$$

Notice in the final expression that a_i has been instantiated using the particular attribute values of the new instance. To calculate v_{NB} we now require 10 probabilities that can be estimated from the training data. First, the probabilities of the different target values can easily be estimated based on their frequencies over the 14 training examples

$$\begin{aligned} P(\text{PlayTennis} = \text{yes}) &= 9/14 = .64 \\ P(\text{PlayTennis} = \text{no}) &= 5/14 = .36 \end{aligned}$$

Similarly, we can estimate the conditional probabilities. For example, those for *Wind = strong* are

$$\begin{aligned} P(\text{Wind} = \text{strong} | \text{PlayTennis} = \text{yes}) &= 3/9 = .33 \\ P(\text{Wind} = \text{strong} | \text{PlayTennis} = \text{no}) &= 3/5 = .60 \end{aligned}$$

Using these probability estimates and similar estimates for the remaining attribute values, we calculate v_{NB} according to Equation (6.21) as follows (now omitting attribute names for brevity)

$$\begin{aligned} P(\text{yes}) P(\text{sunny} | \text{yes}) P(\text{cool} | \text{yes}) P(\text{high} | \text{yes}) P(\text{strong} | \text{yes}) &= .0053 \\ P(\text{no}) P(\text{sunny} | \text{no}) P(\text{cool} | \text{no}) P(\text{high} | \text{no}) P(\text{strong} | \text{no}) &= .0206 \end{aligned}$$

Thus, the naive Bayes classifier assigns the target value *PlayTennis = no* to this new instance, based on the probability estimates learned from the training data. Furthermore, by normalizing the above quantities to sum to one we can calculate the conditional probability that the target value is *no*, given the observed attribute values. For the current example, this probability is $\frac{.0206}{.0206 + .0053} = .795$.

Figure 2.2: Probability Calculation for PlayTennis

2.1.2 Solution

We need to draw the Bayesian belief network for the naive Bayes classifier based on the given data and calculate the conditional probability associated with the node **Wind**.

Part 1: Drawing the Bayesian Belief Network

A naive Bayes classifier assumes that all features (attributes) are conditionally independent given the target variable (class label). In this case, the target variable is **PlayTennis** (with values **yes** or **no**), and the features are **Outlook**, **Temperature**, **Humidity**, and **Wind**.

In a Bayesian belief network for a naive Bayes classifier:

- The target variable (**PlayTennis**) is the parent node.
- Each feature (**Outlook**, **Temperature**, **Humidity**, **Wind**) is a child node of **PlayTennis**.
- No edges between feature nodes because of conditional independence assumption.

So, the structure of the network can be represented as:

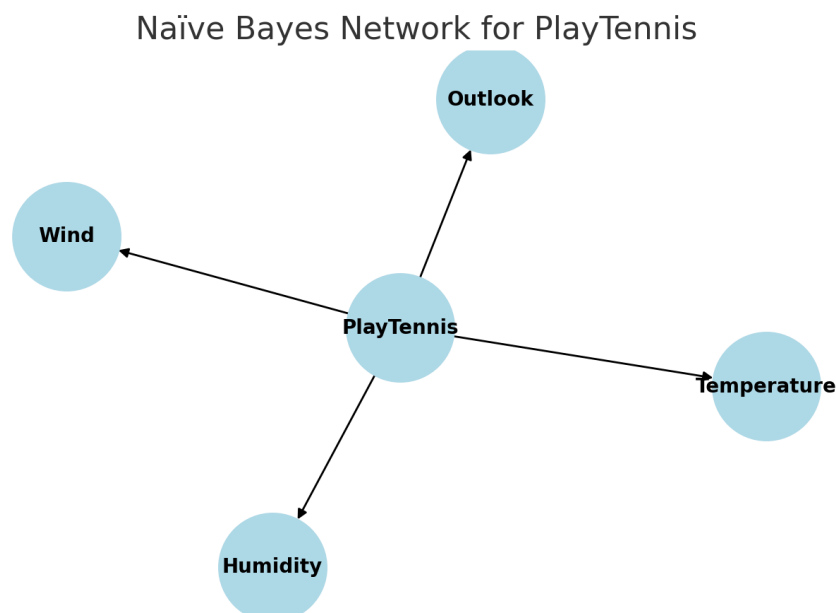


Figure 2.3: Bayesian Belief Network

- **Nodes:** **PlayTennis**, **Outlook**, **Temperature**, **Humidity**, **Wind**.
- **Edges:** Directed edges from **PlayTennis** to each of the feature nodes (**Outlook**, **Temperature**, **Humidity**, **Wind**).
- **No edges between features:** This reflects the naive Bayes assumption that **Outlook**, **Temperature**, **Humidity**, and **Wind** are independent of each other given **PlayTennis**.

Part 2: Calculate the Conditional Probability Associated with the Node Wind

The problem asks for the "conditional probability associated with the node **Wind**." In the context of a naive Bayes classifier, this typically means calculating the conditional probabilities of the **Wind** node given the target variable **PlayTennis**, i.e., $P(\text{Wind} \mid \text{PlayTennis})$. These probabilities are used in the naive Bayes classification process and are estimated from the training data.

The possible values for **Wind** are **weak** and **strong**, and the possible values for **PlayTennis** are **yes** and **no**. We need to compute:

- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{yes})$
- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{no})$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{yes})$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{no})$

These probabilities are calculated by counting frequencies from the training data in Table 3.2 and normalizing them.

Step 1: Count the Total Instances for Each PlayTennis Value From the table:

- Total instances: 14
- **PlayTennis = yes**: 9 instances (D3, D4, D5, D7, D9, D10, D11, D12, D14)
- **PlayTennis = no**: 5 instances (D1, D2, D6, D8, D13)

This matches the probabilities given in the text:

- $P(\text{PlayTennis} = \text{yes}) = 9/14 \approx 0.64$
- $P(\text{PlayTennis} = \text{no}) = 5/14 \approx 0.36$

Step 2: Count the Instances for Wind Given PlayTennis Now, let's count the occurrences of **Wind** values (**weak** or **strong**) for each value of **PlayTennis**.

For PlayTennis = yes (9 instances):

- D3: Wind = weak
- D4: Wind = weak
- D5: Wind = weak
- D7: Wind = strong

- D9: Wind = weak
- D10: Wind = weak
- D11: Wind = strong
- D12: Wind = strong
- D14: Wind = weak
- Wind = weak: 6 instances (D3, D4, D5, D9, D10, D14)
- Wind = strong: 3 instances (D7, D11, D12)

So:

- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{yes}) = 6/9 \approx 0.67$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{yes}) = 3/9 \approx 0.33$

For PlayTennis = no (5 instances):

- D1: Wind = weak
- D2: Wind = strong
- D6: Wind = strong
- D8: Wind = weak
- D13: Wind = strong
- Wind = weak: 2 instances (D1, D8)
- Wind = strong: 3 instances (D2, D6, D13)

So:

- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{no}) = 2/5 = 0.4$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{no}) = 3/5 = 0.6$

Step 3: Verify with the Text The text provides:

- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{yes}) = 3/9 \approx 0.33$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{no}) = 3/5 = 0.6$

These match our calculations. The probabilities for **Wind = weak** can be derived as the complement:

- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{yes}) = 1 - 0.33 = 0.67$
- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{no}) = 1 - 0.6 = 0.4$

Final Answer for Conditional Probabilities The conditional probabilities associated with the node **Wind** are:

- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{yes}) = 0.67$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{yes}) = 0.33$
- $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{no}) = 0.4$
- $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{no}) = 0.6$

Summary

1. **Bayesian Belief Network:** The network has **PlayTennis** as the parent node with directed edges to **Outlook**, **Temperature**, **Humidity**, and **Wind**. There are no edges between the feature nodes due to the conditional independence assumption.
2. **Conditional Probabilities for Wind:**
 - $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{yes}) = 0.67$
 - $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{yes}) = 0.33$
 - $P(\text{Wind} = \text{weak} \mid \text{PlayTennis} = \text{no}) = 0.4$
 - $P(\text{Wind} = \text{strong} \mid \text{PlayTennis} = \text{no}) = 0.6$